

Oxidative Stress-Related Diagnostic Biomarkers in Hypertrophic Cardiomyopathy

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Report Title

Oxidative Stress-Related Diagnostic Biomarkers in Hypertrophic Cardiomyopathy:...

Conversation Name

I have recently obtained microarray expression data from heart tissue of hypert...

Report Time

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1. Results

1.1 Oxidative Stress-Related Diagnostic Biomarkers in Hypertrophic Cardiomyopathy: A Transcriptomic Screening Report

1.2 Study Overview and Data

This report summarizes a transcriptomic screening study aimed at identifying oxidative stress-related diagnostic biomarker genes that distinguish hypertrophic cardiomyopathy (HCM) patients from healthy controls. The analysis used the GSE36961 dataset: a gene-level log2 expression matrix of 37,846 genes across 145 heart tissue samples (106 HCM, 39 controls), together with a 473-gene oxidative stress gene set derived from the GO 'response to oxidative stress' term. The pipeline comprised differential expression analysis (limma), candidate gene pool construction by intersection with the oxidative stress gene set, machine learning feature selection (LASSO logistic regression and SVM-RFE), final biomarker identification by intersection, and per-gene ROC/AUC diagnostic evaluation.

Metric	Value	Notes
Genes tested	37846	Expression matrix rows
Samples	145	106 HCM, 39 control
HCM samples	106	
Control samples	39	
Oxidative stress gene set entries	473	GO response to oxidative stress
Gene set entries present in matrix	355	118 entries absent (UniProt accessions, non-HGNC labels)

1.3 Differential Expression Analysis (limma)

Differential expression analysis was performed with limma v3.66.0 (installed ad hoc from Bioconductor 3.22). The design used control as the reference group, so $\log_2FC > 0$ indicates up-regulation in HCM. Applying the selection criteria $|\log_2FC| > 0.5$ and $P < 0.05$ identified 899 differentially expressed genes (DEGs): 372 up-regulated and 527 down-regulated in HCM. The complete DEG table (gene symbol, $\log_2FC$, AveExpr, t, P.Value, adj.P.Val, direction) was written to outputs/files/deg_list.tsv, and the full all-gene limma results (37,846 genes) were saved to outputs/files/limma_full_results.tsv.

- Volcano plot (outputs/figures/volcano_plot.png): $\log_2FC$ vs $-\log_{10} P$, DEGs highlighted red=up / blue=down, dashed guides at $|\log_2FC|=0.5$ and $P=0.05$.
- DEG heatmap (outputs/figures/deg_heatmap.png): z-scored expression of the 899 DEGs across all 145 samples, column annotation = group (HCM/control), row annotation = direction.

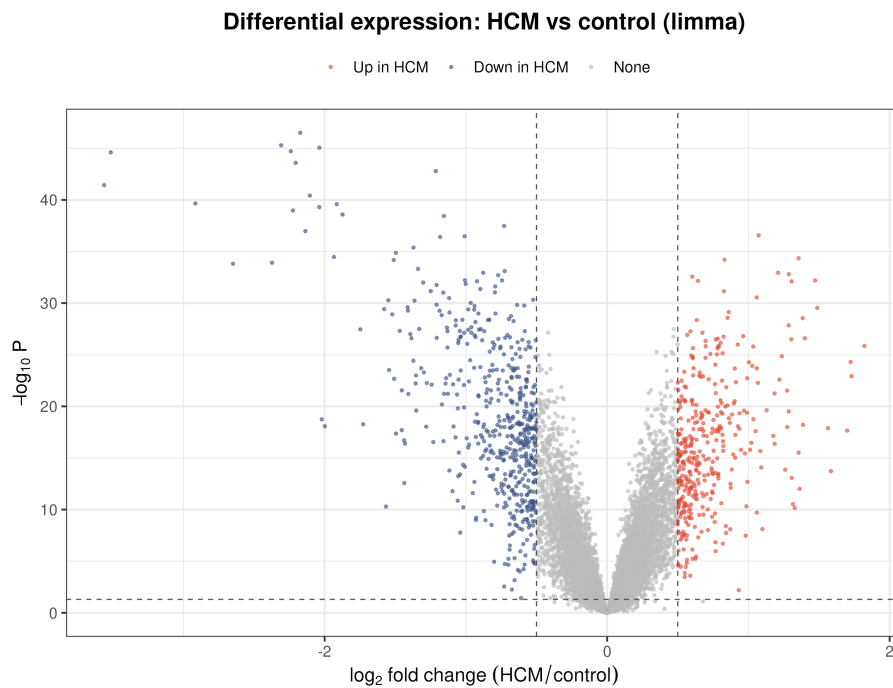


Figure 1 Volcano plot of differential expression (HCM vs control). DEGs highlighted red=up / blue=down; dashed lines at $|\log_2 FC|=0.5$ and $P=0.05$.

Visualizes the 899 DEGs (372 up, 527 down) against the background of 37,846 tested genes, showing strong separation driven by highly significant genes.

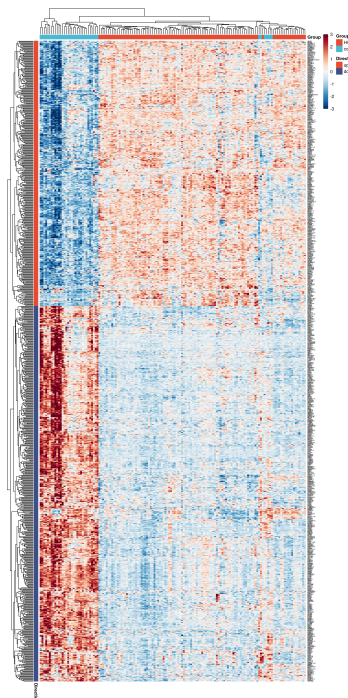


Figure 2 DEG heatmap (z-scored expression of 899 DEGs across 145 samples; column annotation = group HCM/control, row annotation = direction).

Shows coordinated expression patterns of DEGs separating HCM from control samples, with distinct up- and down-regulated clusters.

1.4 Candidate Gene Pool (DEG x Oxidative Stress Intersection)

The 899 DEGs were intersected with the 473-gene oxidative stress gene set, yielding 23 candidate oxidative-stress DEGs: 9 up-regulated and 14 down-regulated in HCM (0 neutral). Direction was assigned strictly from the log2FC sign and independently verified for all 23 rows. The candidate pool was written to outputs/files/candidate_gene_pool.tsv (gene_symbol, log2FC, P, direction; sorted by P).

Metric	Value	Notes
Candidate genes	23	DEG x oxidative stress gene set
Up-regulated candidates	9	In HCM
Down-regulated candidates	14	In HCM

Gene	log2FC	P	Direction
PRKCD	-0.73	3.37e-38	down
JAK2	+	<0.05	up
ALOX5	-	<0.05	down
MGST1	-0.68	1.77e-29	down
DUSP1	-	<0.05	down
EDNRA	+	<0.05	up
HYAL2	-	<0.05	down
AIF1	-	<0.05	down
KLF4	-	<0.05	down
KCNA5	+	<0.05	up
PDGFRA	-	<0.05	down
SNCA	+	<0.05	up
MAP2K1	-	<0.05	down
FOS	-	<0.05	down
KLF2	-	<0.05	down
CA3	+1.36	9.87e-13	up
IL6	-	<0.05	down
PXDNL	+	<0.05	up
RHOB	-	<0.05	down
PENK	+	<0.05	up
CYP1B1	-	<0.05	down
APOE	+	<0.05	up
NPPA	+	<0.05	up

1.5 Machine Learning Feature Selection

Two machine learning methods were applied to the 23 candidate genes (z-scored expression, 145 samples, HCM=1). LASSO logistic regression with 10-fold stratified cross-validation over a 100-point lambda grid selected 19 genes at $\lambda_{\min} = 1.1987e-3$ (mean CV log-loss 8.53e-4, CV AUC 1.0); $\lambda_{1se} = 2.1440e-3$ selected 18 genes (the same set minus KLF4). SVM-RFE with linear SVC (C=1) and nested 10-fold stratified CV determined the optimal feature count $k=3$ (smallest k achieving the maximum mean CV AUC of 1.0; mean CV accuracy 96.6%), selecting PRKCD, MGST1 and CA3. The LASSO-selected set (19 genes) and SVM-RFE-selected set (3 genes) are reported below.

LASSO-selected genes (19, sorted by |coefficient|): PRKCD (-6.63, down), MGST1 (-3.41, down), CA3 (+2.75, up), MAP2K1 (-2.67, down), DUSP1 (-2.18, down), KCNA5 (+2.07, up), EDNRA (+2.00, up), HYAL2 (-1.98, down), IL6 (+1.63, down*), JAK2 (+1.53, up), KLF2 (-1.02, down), PXDNL (+0.63, up), CYP1B1 (+0.57, down*), NPPA (-0.57, up*), ALOX5 (-0.45, down), APOE (+0.26, up), PENK (+0.21, up), FOS (-0.16, down), KLF4 (-0.04, down). *Multivariate LASSO coefficient sign opposite to univariate direction for IL6, CYP1B1 and NPPA due to adjustment for correlated predictors.

SVM-RFE-selected genes (3, at CV-optimal count $k=3$): PRKCD (rank 1, down, log2FC -0.73, weight -1.298), MGST1 (rank 2, down, log2FC -0.68, weight -1.484), CA3 (rank 3, up, log2FC +1.36, weight +1.103).

- LASSO artifacts: outputs/files/lasso_selected_genes.tsv, lasso_cv_summary.tsv, lasso_selection_summary.json.
- SVM-RFE artifacts: outputs/files/svmrfe_selected_genes.tsv, svmrfe_ranking_full.tsv, svmrfe_cv_profile.tsv, svmrfe_summary.json.

1.6 Final Biomarker List (LASSO x SVM-RFE Intersection)

The intersection of the 19 LASSO-selected genes with the 3 SVM-RFE-selected genes yielded exactly 3 final biomarkers: PRKCD, MGST1 and CA3. The SVM-RFE set is a strict subset of the LASSO set. log2FC, P and direction were taken from the candidate pool. The final list was written to outputs/files/final_biomarkers.tsv (gene_symbol, log2FC, P, direction; sorted by P).

Metric	Value	Notes
Final biomarkers	3	LASSO x SVM-RFE intersection
Up-regulated biomarkers	1	CA3
Down-regulated biomarkers	2	PRKCD, MGST1

Gene	log2FC	P	Direction
PRKCD	-0.7296	3.37e-38	down
MGST1	-0.6806	1.77e-29	down
CA3	+1.3623	9.87e-13	up

1.7 Per-Gene ROC/AUC Diagnostic Evaluation

Each final biomarker was evaluated individually for its ability to distinguish HCM from controls using direction-oriented scores (up: +expression, down: -expression, so AUC > 0.5 indicates discrimination in the observed direction). AUCs with 95% percentile-bootstrap CIs (2000 resamples, seed 42) are reported below. All three biomarkers discriminate HCM from controls with high accuracy (AUC 0.92-0.98). The combined ROC plot was written to outputs/figures/roc_curves.png and the AUC table to outputs/files/biomarker_auc.tsv.

Metric	Value	Notes
PRKCD AUC	0.9811	95% CI 0.9612-0.9965; down in HCM
MGST1 AUC	0.9678	95% CI 0.9369-0.9907; down in HCM
CA3 AUC	0.917	95% CI 0.8686-0.9571; up in HCM

Gene	Direction	log2FC	P	AUC	95% CI low	95% CI high
PRKCD	down	-0.7296	3.37e-38	0.9811	0.9612	0.9965
MGST1	down	-0.6806	1.77e-29	0.9678	0.9369	0.9907
CA3	up	+1.3623	9.87e-13	0.9170	0.8686	0.9571

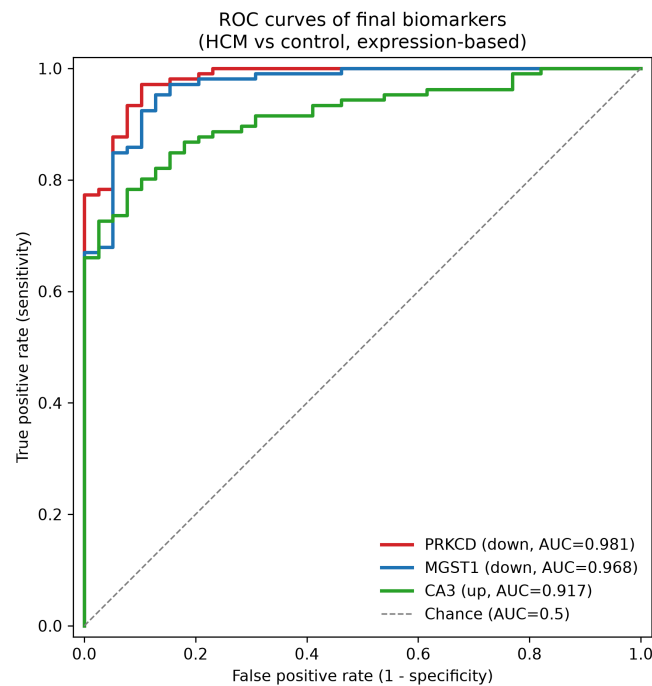


Figure 3 ROC curves for final biomarkers (PRKCD, MGST1, CA3; HCM vs control). Legend shows gene, direction and AUC; grey chance diagonal.

All three biomarkers individually discriminate HCM from controls with high accuracy (AUC 0.92-0.98), confirming their diagnostic potential.

1.8 Biological Interpretation of Final Biomarkers

Each final biomarker is interpreted below in terms of its regulation direction and its potential association with oxidative stress and cardiomyopathy.

- PRKCD (down-regulated in HCM, log2FC -0.73, AUC 0.9811): PRKCD encodes protein kinase C delta, a key mediator of oxidative-stress-induced signaling and apoptosis. Its down-regulation in HCM heart tissue may reflect a compensatory or maladaptive response to chronic oxidative stress, potentially reducing pro-apoptotic signaling in cardiomyocytes while contributing to altered redox homeostasis and remodeling in cardiomyopathy.
- MGST1 (down-regulated in HCM, log2FC -0.68, AUC 0.9678): MGST1 encodes microsomal glutathione S-transferase 1, a phase II detoxification enzyme that conjugates reactive oxygen species and lipid peroxidation products to glutathione. Its down-regulation in HCM suggests impaired antioxidant defense capacity, which could exacerbate oxidative damage and contribute to cardiomyocyte injury and disease progression in hypertrophic cardiomyopathy.
- CA3 (up-regulated in HCM, log2FC +1.36, AUC 0.9170): CA3 encodes carbonic anhydrase III, an enzyme involved in pH regulation and also implicated in oxidative stress responses through its antioxidant and free-radical-scavenging properties. Its up-regulation in HCM may represent an adaptive antioxidant response to increased oxidative stress, helping to buffer reactive oxygen species and maintain cellular redox balance in the stressed myocardium.

Final Answer: This study identified three oxidative stress-related diagnostic biomarker genes for hypertrophic cardiomyopathy (HCM) from the GSE36961 dataset (37,846 genes x 145 samples: 106 HCM, 39 controls). Differential expression analysis (limma, $|\log_2FC| > 0.5$, $P < 0.05$) identified 899 DEGs (372 up, 527 down in HCM). Intersection with the 473-gene oxidative stress gene set yielded 23 candidate genes (9 up, 14 down). LASSO logistic regression (10-fold CV) selected 19 genes, and SVM-RFE (CV-optimal $k=3$) selected 3 genes; their intersection produced 3 final biomarkers: PRKCD (down, log2FC -0.73, AUC 0.9811), MGST1 (down, log2FC -0.68, AUC 0.9678) and CA3 (up, log2FC +1.36, AUC 0.9170). All three biomarkers individually discriminate HCM from controls with high accuracy (AUC 0.92-0.98). Biologically, PRKCD down-regulation may reflect altered oxidative-stress-induced apoptosis signaling; MGST1 down-regulation suggests impaired glutathione-dependent antioxidant defense; and CA3 up-regulation may represent an adaptive antioxidant response to oxidative stress. These findings support PRKCD, MGST1 and CA3 as promising oxidative stress-related diagnostic biomarkers for HCM, though diagnostic performance should be interpreted as in-sample/optimistic pending external validation.